



ESRI
EXTERNAL

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DHIS2 to ArcGIS Connector

DHIS2 to ArcGIS Connector App v0.1 and
DHIS2 Custom Data Feed v0.1
User Manual

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Introduction

Background

In recent years, geospatial data and visualization have become integral to analysis, decision-making, and communication. The **DHIS2 to ArcGIS Connector App**—developed using the DHIS2 app development framework—enables users to seamlessly integrate their DHIS2 data into the ArcGIS system through a live, no-code connection. Available in the [DHIS2 App Hub](#), this app bridges the gap between DHIS2 data management and the powerful spatial analytics and visualizations in the ArcGIS System.

The app addresses a critical need for interoperable systems that enhance the value of data. It augments the capabilities of the DHIS2 Maps app and unlocks new opportunities for data storytelling, decision support, and stakeholder engagement.

For additional information on ArcGIS access, DHIS2 users can explore Esri's offerings [here](#).

This app is an open-source project powered by Esri but is not a supported Esri Product. Learn more about contributing to the project at the [Esri GitHub for the DHIS2 to ArcGIS Connector App](#) and [Esri GitHub for the DHIS2 Custom Data Feed](#).

Prerequisites

Before you begin, ensure you have:

- A **DHIS2 instance** (2.37 or higher recommended)
- An **ArcGIS Enterprise account** (11.3 or higher recommended) with Custom Data Feed Runtime configured on the hosting server
 - Note: If you have an ArcGIS Enterprise instance that does not have Custom Data Feeds configured, follow [these instructions](#) to configure them on your hosting server (administrative access required)
- Setting up the DHIS2 to ArcGIS Connector app requires admin access to the ArcGIS Enterprise instance. Once the app has been set up and configured, all users in the DHIS2 – ArcGIS Enterprise environments will have access to use the connector app.

DHIS2

DHIS2 is a free and open-source platform for collecting, reporting, analyzing and disseminating aggregate and individual-level data. Most commonly used for health data, DHIS2 can be implemented for individual health programs or as a national-scale health management information system. Officially recognized as a Digital Public Good, DHIS2 is also used in sectors such as education, logistics, sanitation, and land management, among others.

DHIS2 is widely used globally, particularly in low-resource settings, to strengthen health and other information systems and improve data-driven decision-making. Its open-source nature, flexibility, and scalability make it a valuable tool for managing data and improving outcomes. Looking ahead, the use cases of DHIS2 are expanding to education, land administration, and logistics. For full DHIS API Documentation and use please visit: (<https://docs.dhis2.org/en/home.html>)

The ArcGIS System

ArcGIS is a comprehensive geographic information system (GIS) developed by Esri that enables users to create, manage, analyze, and share spatial data. ArcGIS delivers industry-leading mapping and analytics to your infrastructure and the cloud. It provides powerful tools for mapping and visualizing data, uncovering patterns, and making data-driven decisions across sectors such as health, environment, urban planning, logistics, and more.

ArcGIS includes a suite of products that work across desktop, web, mobile, and enterprise environments—such as ArcGIS Online, ArcGIS Pro, and ArcGIS Enterprise—allowing users to build interactive maps, dashboards, and geospatial applications. Whether used for field data collection, real-time monitoring, or predictive modeling, ArcGIS supports evidence-based planning and collaboration with location intelligence at its core.

ArcGIS Enterprise

ArcGIS Enterprise is Esri's comprehensive geospatial platform designed for organizations that need to manage, analyze, and share spatial data securely within their own IT environment. It includes powerful tools for mapping, data visualization, spatial analytics, and collaboration. Built on a scalable architecture, ArcGIS Enterprise supports advanced

workflows, real-time data integration, and customizable apps—all while maintaining control over data privacy, infrastructure, and user access.

This app allows DHIS2 users to leverage their data in the ArcGIS system using a direct connection to an ArcGIS Enterprise instance that maintains data hosting in their DHIS2 instance. The app uses a no-code interface to allow for seamless interoperability. The connection is formed through [custom data feeds](#). View the GitHub repository for the [DHIS2 Custom Data Feeds] (<https://github.com/ArcGIS/dhis2-custom-data-feeds>) for more information.

Privacy and Security

The ArcGIS Connector uses OAuth 2.0 for secure user authentication with your ArcGIS Enterprise deployment. For more information, please see the [documentation](#). Access to data within the app respects the user permissions defined in DHIS2, ensuring users only see what they're authorized to view. The connector establishes a live, authenticated data feed from DHIS2 to the ArcGIS system—without duplicating or storing the data externally. Unless explicitly exported or copied, all data remains securely within the DHIS2 instance. Any changes made—such as edits or deletions—are automatically reflected in ArcGIS, preserving data integrity and synchronization.

Setting up the DHIS2 Custom Data Feed in ArcGIS Enterprise

Install Custom Data Feeds

To use the dhis2 CDF providers, you must install Custom Data Feeds onto your ArcGIS Server Machine and install the ArcGIS Enterprise SDK on your development machine. Your development machine can be your own laptop.

In general, you install Custom Data Feeds by:

1. Install the ArcGIS Server Custom Data Feeds Runtime on each machine in your ArcGIS Server site.
2. Ensure Node.js is installed to C:\Program Files\nodejs *before* installing the ArcGIS Enterprise SDK. The SDK checks for a supported Node.js version during installation, and the CDF CLI may fail to activate if Node.js is missing or installed elsewhere.

- Ensure that the version of Node.js on your machine is compatible with your version of ArcGIS Enterprise.
 1. [Refer to the documentation for supported Node.js versions](#)
 2. In a terminal, you can check your node version with the command
`node -v`
- 3. Install the ArcGIS Enterprise SDK on your development machine (note: this does not need to be the same machine where the server is, but can be).
 - When running the ArcGIS Enterprise SDK installer, select the optional Custom Data Feeds component when prompted by the installer (the CDF CLI is not installed by default). After installation, open a terminal and run `cdf -h` to confirm the CLI is available.
 - If you see a "command not found" error, run the activation script (`activate_cdf.bat` on Windows or `activate_cdf.sh` on Linux), located in the `customdatacli` directory of the SDK install (e.g. `C:\Program Files\ArcGIS\EnterpriseSDK\customdatacli`), then re-open the terminal and try again.

Complete instructions for installing Custom Data Feeds are available in the [ArcGIS Developer Documentation](#).

Configure and Publish Your Provider

After installing the ArcGIS Server Custom Data Feeds runtime and setting up your development machine, you must configure your providers.

1. **Create your Custom Data App on your Development Machine**
 1. Open a terminal and navigate to the directory where you will configure your dhis2 Custom Data Feed providers.
 2. Run the command `cdf createapp <app-name>`
2. **Install the dhis2 Custom Data Feed Providers**
 1. In your terminal, navigate to the new directory created by the CDF CLI. This directory should match what you passed as `<app-name>`
 2. Clone the repo directly into a folder named `providers`:
`git clone https://github.com/ArcGIS/dhis2-custom-data-feeds.git providers`

- Your providers directory should now contain a subdirectory for each dhis2 cdf provider.
3. In your terminal, navigate to each provider directory (e.g. `analytics`) and run the command `npm install`.

Configure the Providers

2. Open your CDF App directory in your IDE of choice
3. Go the `config` directory create a *new file* named `default.json`
4. Open `default.json` and add the following:

```
{ "DHIS2_KEY": "YOUR_ACCESS_TOKEN", "DHIS2_API":  
  "DHIS2_API_URL" }
```

Be sure to set `DHIS2_API` as the full base url to your dhis2 instance's Web API e.g. <https://<your-dhis2-instance>/dhis/api/>

If you don't know how to get a DHIS2 Access Token, see below. If you already have it, skip to Publish the Provider.

Note: This example uses `/api/` without a version number, which will typically default to your instance's current supported Web API version. You can optionally pin a specific version by appending the version number (e.g. `/api/40`, which corresponds to the DHIS2 2.40 release line). If you do pin a version, confirm your instance actually supports it, or the data feed may fail to return data. You can check your instance's version by visiting `https://<your-dhis2-instance>/api/system/info` in a browser.

Generating a DHIS2 Access Token

The `DHIS2_KEY` value in `default.json` is a Personal Access Token (PAT) generated from a DHIS2 account. The token inherits all the permissions and authorities of the user who creates it, so for a server deployment it is recommended to create a dedicated service account with only the roles needed to read the data the feed requires, rather than using a personal account.

Note: A Personal Access Token is recommended (read-only scope and a controlled expiry), but the `DHIS2_KEY` field also accepts a DHIS2 account's credentials via Basic Authentication, the same type of credential used by integrations such as Power BI. If you use account credentials, note that they grant that account's full permissions and that the feed will break if the password is later rotated or expires. Either credential is stored in plain text in `default.json`, so prefer a dedicated, minimally-scoped service account.

To generate a token:

1. Confirm Personal Access Tokens are enabled on the instance. PATs require `enable.api_token.authentication` to be set to `on` in the server's `dhis.conf` file, and the feature is disabled by default before DHIS2 2.38. If the “Personal access tokens” menu does not appear, this setting is likely the cause.
2. In DHIS2, click your user avatar in the top-right corner and choose **Edit profile**.
3. Choose **Personal access tokens** from the left-hand navigation sidebar, then click **Generate new token**.
4. Select the server/script context, which applies to integrations and scripts that are not accessed by a browser. This context also allows you to restrict the set of IP addresses from which the token may be used (e.g. the ArcGIS Server's IP).
5. Restrict the token to the GET HTTP method only. The connector is read-only and does not need to modify or delete data.
6. Set the expiration deliberately. The default is 30 days, after which the token returns a 401 (Unauthorized) error and the data feed will stop working. Record the renewal date.
7. Copy the token immediately. The token key is shown only once—the server stores only a hash. If it is lost, you must generate a new one. The key begins with `d2pat_`.
8. Paste the full token string into the `DHIS2_KEY` field in `default.json`.

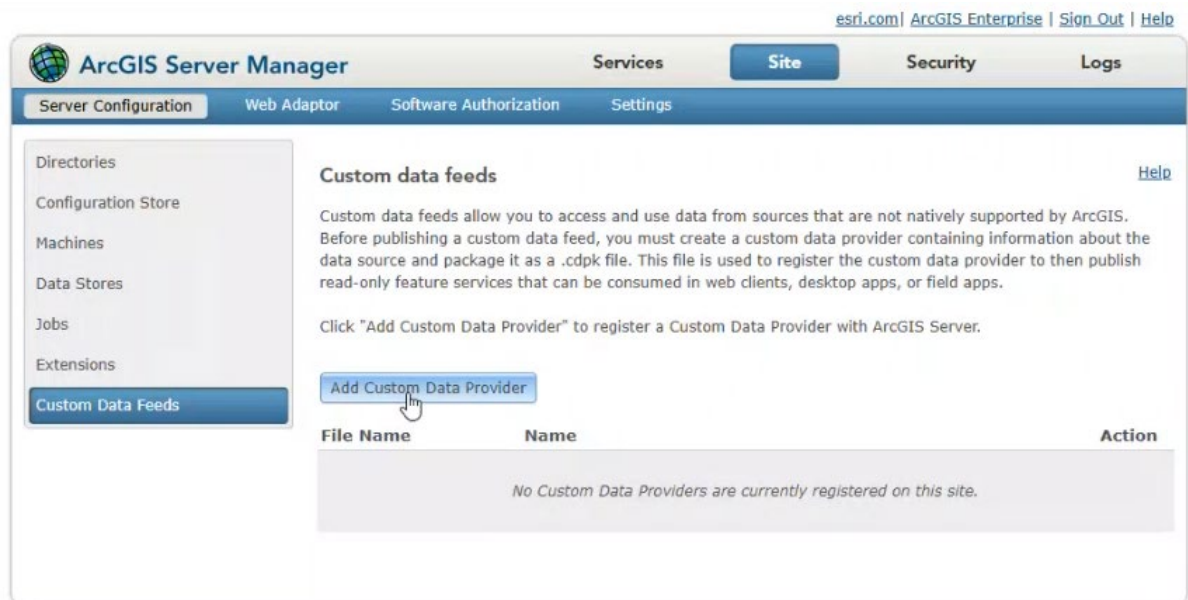
Package the Provider

1. In your terminal, stop the node server if it is running.
2. In your terminal, navigate to the directory created by the `CDF createapp` command
3. Run the command `cd export analytics`
4. You should now see a file named `analytics.cdpr`

Note: If `cdf export` reports that the provider does not exist, check that you are running the command from the app directory created by `cdf createapp` (not from inside `providers` or a provider folder), and that the provider directory (e.g. `analytics`) sits directly inside the `providers` folder. If you instead see the provider nested inside an extra `dhis2-custom-data-feeds` folder, the repository was cloned one level too deep; move the provider folders up so they sit directly inside `providers` and run the command again. Also confirm you are passing the provider *name* (e.g. `analytics`), not a file path.

Register Your Custom Data Feed Provider

1. Locate the `.cdpk` file you generated in the Publish the Provider step (e.g. `analytics.cdpk`). This is the file you will register—you do not need to download anything separately from GitHub.
2. In the browser, navigate to `https://[arcgisenterprisename].com/server/manager`
3. Log in with the administrator credentials.
4. Select **Site**.
5. Select **Server Configuration**.
6. Navigate to **Custom Data Feeds**.
7. Select **Add Custom Data Provider**.



8. Add the .cdpk file you created in step 4 of [Publish the Provider](#)



9. The Custom Data Provider should now connect directly to DHIS2 once you add the app and configure the connection.

Configuring Multiple DHIS2 Instances

A single ArcGIS Enterprise deployment can host providers for more than one DHIS2 instance. Because each provider is registered by its name, every instance requires its own provider with a unique name and its own configuration. Registering two providers with the same name will cause a conflict, so the key is to create a separate, distinctly named copy of the provider for each DHIS2 instance.

To add an additional DHIS2 instance:

1. In your `providers` directory, make a copy of the analytics provider folder and give it a unique name (e.g. `analytics-instanceB`).
2. In the copied folder, open `cdconfig.json` and change the "name" field to match the new folder name (e.g. `"name": "analytics-instanceB"`). This is the value that identifies the provider when it is registered in ArcGIS Enterprise. You do not need to edit `index.js` or `model.js`; they read the name from `cdconfig.json` automatically.
3. In the copied folder, open `config/default.json` and set `DHIS2_API` and `DHIS2_KEY` to the URL and access token for the additional DHIS2 instance. Each provider copy reads its own `default.json`, so the instances remain fully independent.
4. From within the copied provider folder, run `npm install`.

5. From the app directory created by `cdf createapp`, export the new provider using its name: `cdf export analytics-instanceB`. This produces `analytics-instanceB.cdpk`.
6. Register the new `.cdpk` in ArcGIS Server Manager following the same steps in **Register Your Custom Data Feed Provider**.
7. Repeat these steps for each additional DHIS2 instance, giving each a unique provider name and its own `default.json`.

Note: Keep the provider folder name, the "name" field in `cdconfig.json`, and the name you pass to `cdf export` consistent for each provider. Aligning all three avoids confusion and ensures the correct provider is exported and registered.

Troubleshooting and Common Issues

“cdf: command not found”

If the `cdf` command is not recognized even though the ArcGIS Enterprise SDK is installed, the CLI was either not activated or not installed. Try the following in order:

1. Run the activation script manually: `activate_cdf.bat` (Windows) or `activate_cdf.sh` (Linux). It is located in the `customdatacli` directory of the SDK install (e.g. `C:\Program Files\ArcGIS\EnterpriseSDK\customdatacli`). Re-open the terminal and run `cdf -h` again.
2. If that does not resolve it, the cause is almost always Node.js install order. Node.js must be installed before the SDK. Uninstall the ArcGIS Enterprise SDK, install a supported Node.js version, then reinstall the SDK.
3. Install Node.js using only one method—either the direct installer or NVM, never both. Mixing the two is a known cause of CLI issues. On Windows, Node.js should be installed to `C:\Program Files\nodejs`.
4. On Linux, ensure the same user installs both Node.js and the CDF CLI.
5. If the activation script does not exist at all, the Custom Data Feeds component was not selected during SDK installation. Re-run the installer and ensure Custom Data Feeds is selected.

Node.js version compatibility

The Custom Data Feeds CLI and Runtime are sensitive to the Node.js version. Run `npm install` and `cdf export` using a Node.js version that matches the one running on your ArcGIS Server. Always confirm the supported version against the current ArcGIS Enterprise SDK system requirements for your Enterprise release rather than assuming.

“Provider does not exist” when running `cdf export`

This usually means the repository was cloned into a subfolder instead of directly into the `providers` directory. The clone command ends with a space and a period (`git clone https://github.com/ArcGIS/dhis2-custom-data-feeds.git .`). The trailing period clones the contents into the current directory; without it, `git` creates an extra `dhis2-custom-data-feeds` subfolder and the provider directories end up one level too deep. The provider folders (e.g. `analytics`) must sit directly inside `providers`. Also note that `cdf export` takes the provider name (e.g. `analytics`), not a file path.

Provider not appearing after registration

The Custom Data Feeds Runtime is a separate install from the SDK—the SDK and CLI go on the development machine, while the Runtime must be installed on the ArcGIS Server machine itself. For ArcGIS Enterprise 11.x, the Runtime must be installed on every machine in the Server site; missing a machine in a multi-machine deployment causes intermittent failures. If a freshly installed provider does not appear as available, restarting the ArcGIS Server service on the affected machine(s) is the first thing to try.

Updating or renaming a provider

Do not rename a `.cdpk` file or hand-edit its contents to change a provider’s name. The provider name is set when the provider is created and is written inside the package. To change configuration values such as the token or API URL, edit `default.json` on the development machine and re-run `cdf export`. When updating an already-registered provider, the provider name must match the previously registered name, or ArcGIS Server will not recognize it as an update. For this reason, keep the development-machine project folder intact after deployment in case a value needs to be corrected and the package rebuilt.

Token storage

The DHIS2 access token is stored in plaintext in `default.json` on the development machine and is bundled into the `.cdpk`. Generate the token from a dedicated, minimally-scoped service account rather than a personal account, and be deliberate about where the token-bearing project folder and package are kept.

Using the DHIS2 to ArcGIS Connector App

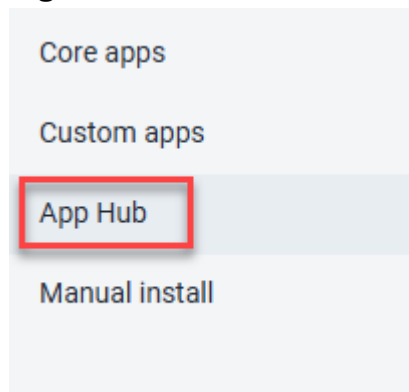
Install the App

1. Log in to your DHIS2 Training Instance.
2. Navigate to **App Management**.



App
Management

3. Open the **DHIS2 App Management**.



4. Navigate to the **App Hub**.
5. In “Search AppHub Apps” search for the **DHIS2 to ArcGIS Connector app** and add it.
6. Once added, you will find it in your list of DHIS2 apps.

Making the App Visible to Other Users

After the app is installed, it may only be visible to administrators. In DHIS2, each app is gated by an authority, and a user will only see the app if one of their assigned user roles has

that authority enabled. To make the connector available to other users, an administrator should:

1. Open the **Users** app.
2. Go to **User Role** management and edit the role (or roles) assigned to the users who need access.
3. Scroll to the **Authorities** section and check the box for the DHIS2 to ArcGIS Connector app.
4. Click **Save**.
5. Have an affected user refresh or clear their app cache and sign in again to confirm the app now appears.

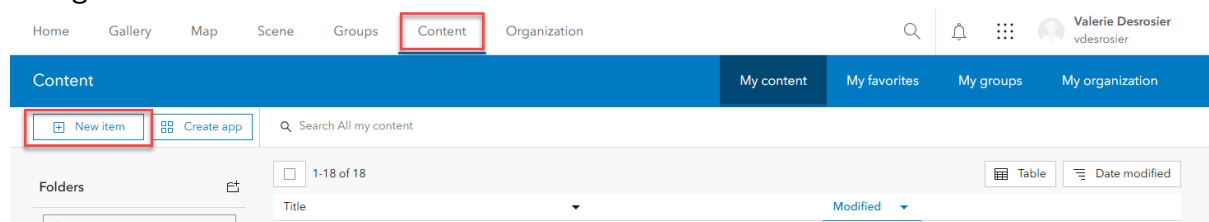
Note: Verify that the users are actually assigned the role you edited. If the connector's authority does not appear in the Authorities list at all, this points to an app installation or sharing issue rather than a role configuration issue.

Configure the DHIS2 and ArcGIS Connection



DHIS2 ArcGIS
Connector

1. Open the **DHIS2 to ArcGIS Connector** app.
2. On the first launch, you will be prompted to configure the connection.
3. The **configuration page** provides documentation and setup instructions.
4. Open a new tab in the same browser and log in to your **ArcGIS Enterprise** account.
5. Navigate to **Content** and select **Create a New Item**.



6. Choose **Application** as the item type.

New item ⓘ



Drag and drop your file or choose an option

 Your device



Feature layer
Create an editable layer with fields copied from a template or feature layer.



Developer credentials
Create API key and OAuth 2.0 credentials to build custom applications.



Tile layer
Create a fast drawing vector tile layer, 3D tiles layer, or raster tile layer.



URL
Link to an ArcGIS Server web service, CSV, OGC web service, KML, GeoJSON or a document.



Application
Link to an application on the web or create a new application.



Scene layer
Create a fast drawing scene layer using 3D content.

7. Select **Other Application**, then click **Next**.

New item ⓘ

Application type

☐ Web mapping

☐ Desktop

☐ Mobile

☒ Other application

8. Name the application with a unique title (e.g., “DHIS2 Connection App”).

9. Select a folder, add tags, and provide a summary.

New item

Title

DHIS2 Connection App

Folder

DHIS2 Connection Folder

Tags

DHIS2

Add tags

Summary

This is the connection app for the DHIS2 to ArcGIS Connector. DO NOT DELETE.

Characters left: 1972

10. Click **Save**.

Configure the OAuth Credentials

1. After creating the application, open it in your content.
2. Locate the **Credentials** section on the item page. Click **Manage**.

HomeGalleryMapSceneGroupsContentOrganization

DHIS2 Connection App

OverviewSettings

Edit thumbnail



Add to Favorites

DHIS2 Connection App - Do not delete.

Application by vdesrosier

Item created: Feb 23, 2025 Item updated: Feb 23, 2025 View count: 0

OAuth 2.0 Credentials

Description

Add an in-depth description of the item.

Credentials

You can use OAuth 2.0 to authenticate ArcGIS users to access content and location services on behalf of the authenticated user.

Client ID

qEU8BfMkc7MekoP

Client Secret

.....

Temporary Token

ll-79lcCGC4eqe8KIO5mYs9PeGwgtpig6fGITTIN0gKkgsclmVMhEl_2UKzXg2VFJwRSCP3ITfPqyBbeeVDdNhUi7tBxkMCafrQsF-DUhw...

Short lived token for testing. Will expire in 2 hour(s) (Thursday, March 6, 2025 at 6:21:22 PM).

View

Share

Item Information

LowHigh

Top Improvement: Add a longer summary

Details

Size: 0 KB

ID: fa3e25e919a0416b95ccad177fc04636

☆☆☆☆

Share

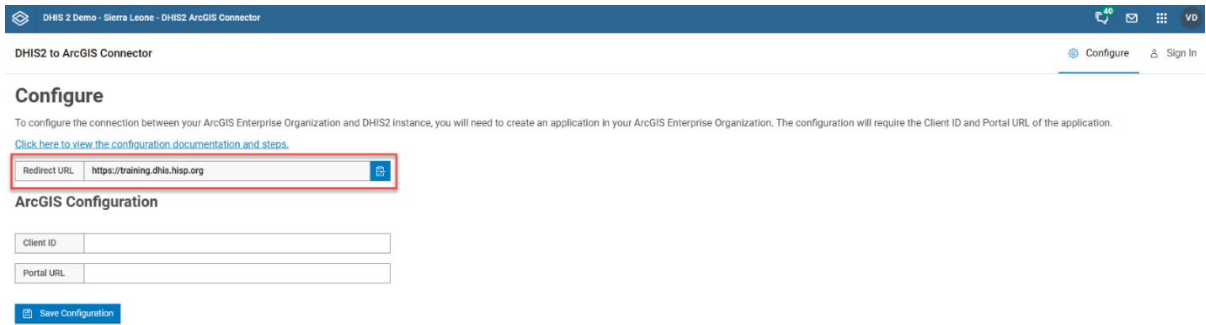
Edit

Owner

vdesrosier

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3. On your DHIS2 tab, copy the **redirect URL** from the DHIS2 configuration page.



DHIS2 to ArcGIS Connector

[Configure](#) [Sign In](#)

Configure

To configure the connection between your ArcGIS Enterprise Organization and DHIS2 Instance, you will need to create an application in your ArcGIS Enterprise Organization. The configuration will require the Client ID and Portal URL of the application.

[Click here to view the configuration documentation and steps.](#)

Redirect URL

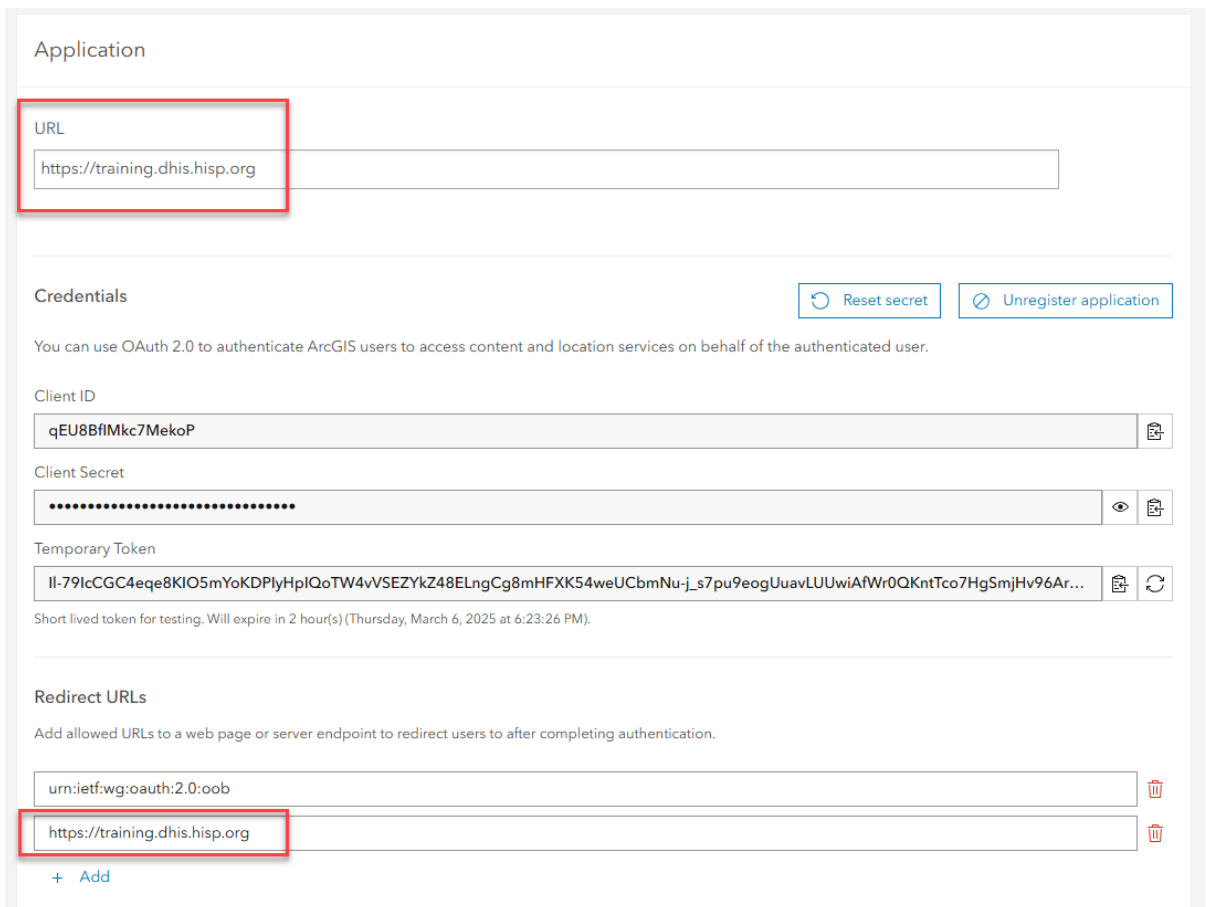
ArcGIS Configuration

Client ID

Portal URL

[Save Configuration](#)

4. Return to the ArcGIS tab. Paste this URL into the **Application URL** and **Redirect URL** fields in ArcGIS.



Application

URL

Credentials

[Reset secret](#) [Unregister application](#)

You can use OAuth 2.0 to authenticate ArcGIS users to access content and location services on behalf of the authenticated user.

Client ID

Client Secret

Temporary Token

Short lived token for testing. Will expire in 2 hour(s) (Thursday, March 6, 2025 at 6:23:26 PM).

Redirect URLs

Add allowed URLs to a web page or server endpoint to redirect users to after completing authentication.

[Delete](#)

[Delete](#)

[+ Add](#)

5. Click **Save**.

6. Copy the **Client ID** from ArcGIS.

Application

URL

Credentials Reset secret Unregister application

You can use OAuth 2.0 to authenticate ArcGIS users to access content and location services on behalf of the authenticated user.

Client ID

Client Secret

Temporary Token

Short lived token for testing. Will expire in 2 hour(s) (Thursday, March 6, 2025 at 6:23:26 PM).

Redirect URLs

Add allowed URLs to a web page or server endpoint to redirect users to after completing authentication.

[+ Add](#)

7. Return to the DHIS2 window, paste it into the DHIS2 Connector configuration.

DHIS 2 Demo - Sierra Leone - DHIS2 ArcGIS Connector

DHIS2 to ArcGIS Connector Configure Sign In

Configure

To configure the connection between your ArcGIS Enterprise Organization and DHIS2 instance, you will need to create an application in your ArcGIS Enterprise Organization. The configuration will require the Client ID and Portal URL of the application.

[Click here to view the configuration documentation and steps.](#)

Redirect URL

ArcGIS Configuration

Client ID

Portal URL

[Save Configuration](#)

8. Copy the **Portal URL** from ArcGIS (including /portal at the end).

Home Gallery Map Scene Groups Content Organization

DHIS2 Connection App Overview Settings

General Application

9. Paste it into the DHIS2 configuration page.

DHIS2 to ArcGIS Connector

Configure

To configure the connection between your ArcGIS Enterprise Organization and DHIS2 instance, you will need to create an application in your ArcGIS Enterprise Organization. The configuration will require the Client ID and Portal URL of the application.
[Click here to view the configuration documentation and steps.](#)

Redirect URL:

ArcGIS Configuration

Client ID:

Portal URL:

[Save Configuration](#)

10. Click **Save Configuration** to update the settings.

11. After saving, go to the **sign-in** page.

DHIS2 to ArcGIS Connector

Configure Sign In

To configure the connection between your ArcGIS Enterprise Organization and DHIS2 instance, you will need to create an application in your ArcGIS Enterprise Organization. The configuration will require the Client ID and Portal URL of the application.
[Click here to view the configuration documentation and steps.](#)

Redirect URL:

ArcGIS Configuration

Client ID:

Portal URL:

[Save Configuration](#)

12. Allow permissions when prompted by ArcGIS.

Request for Permission

esri

vdesrosier

[Sign in with another account](#)

DHIS2 Connection App (Developed by DHIS2 ArcGIS Enterprise 11.4) wants to access your ArcGIS Enterprise account information

Cancel Allow

13. If already logged in, the system will detect your account automatically.

14. If needed, sign in with a different ArcGIS account.

How to Create a Data Connection

1. In DHIS2, open the **DHIS2 to ArcGIS Connector** app.

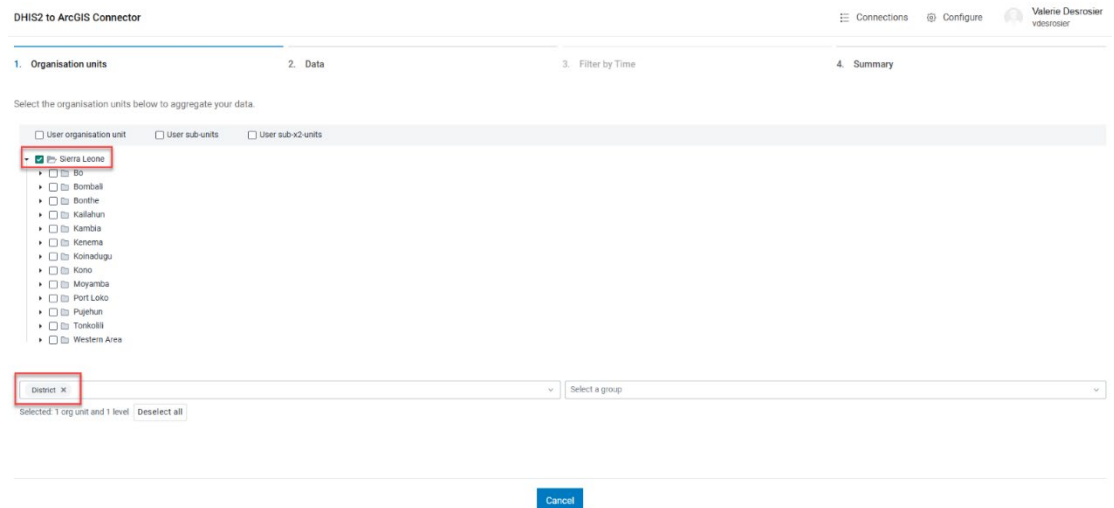


2. If no existing connections are available, click **Add a New Connection**.

3. In the connection setup window:

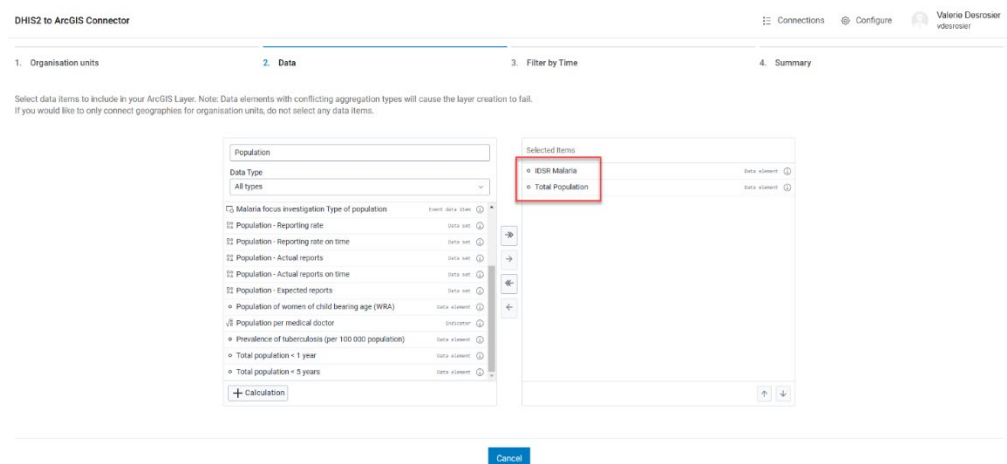
- a. Step 1. Organisation Units

- i. **Select Organization Units** (e.g., Sierra Leone → Districts)



- b. **Select Step 2. Data**

- i. **Choose Data Elements** (e.g., IDSR Malaria, Population).



- c. **Select Step 3. Filter by Time**

i. **Specify the Time Range** (e.g., Last 12 Months).

DHIS2 to ArcGIS Connector

1. Organisation units 2. Data 3. Filter by Time 4. Summary

Select the time period for your selected data. Data may be additionally filtered by time in ArcGIS Enterprise applications and maps.

Relative periods Fixed periods

Period type: Months

- This month
- Last month
- Last 3 months
- Last 6 months
- Months this year

Selected Periods

- Last 12 months

Cancel

4. **Select Step 4. Summary**

a. In the summary window:

- Provide a **unique layer name** for ArcGIS. (Note: best practices are to not include spaces or special characters)
- Provide a **Layer Description** for the data.

DHIS2 to ArcGIS Connector

1. Organisation units 2. Data 3. Filter by Time 4. Summary

Create a title and description for your new ArcGIS Enterprise Layer. Both fields may be changed later in ArcGIS Enterprise.

Layer Name: DHIS2_MalariaCases_IDSR

Layer Description: DHIS2 Malaria Cases and Population by District - Last 12 Months

Create Connection

iii. Click **Create Connection**.

Verify and Access Data in ArcGIS Enterprise

- Once the connection is created, if needed, refresh the connection list in DHIS2. The resulting layer will appear at the top of the list.
- Select **View in ArcGIS – Open**.

DHIS2 to ArcGIS Connector

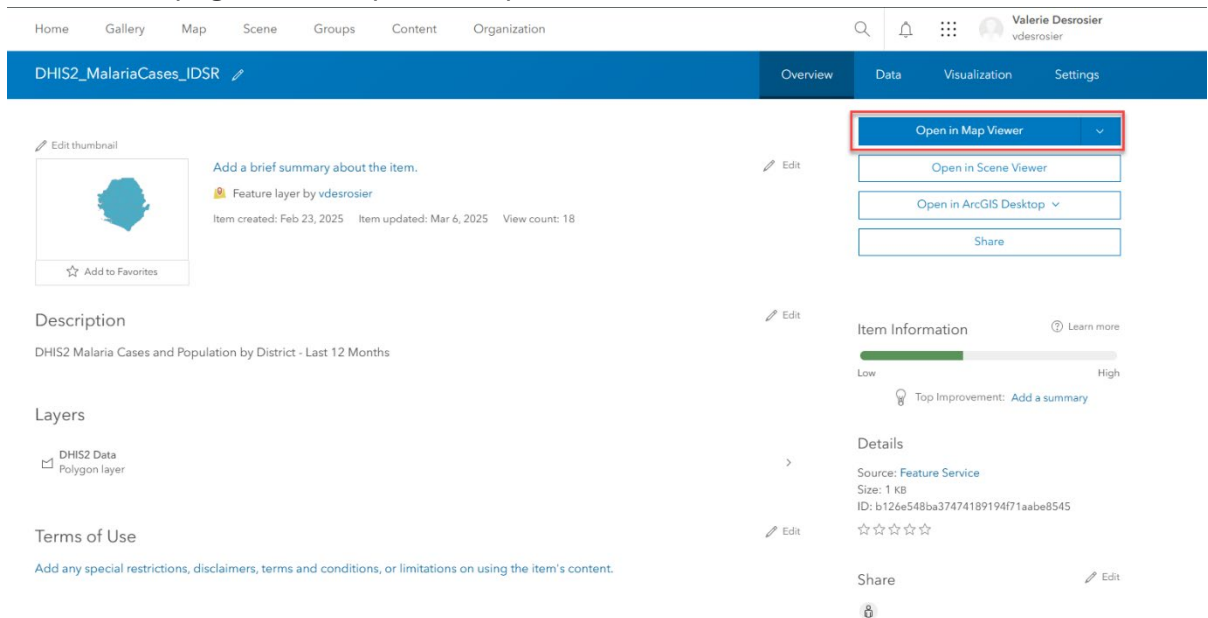
Connections Add New Connection

Title	Description	Created By	Created On	View in ArcGIS
DHIS2_MalariaCases_IDSR	DHIS2 Malaria Cases and Population by District - Last 12 Months	vdesrosier	2/23/2025, 6:04:10 PM	Open

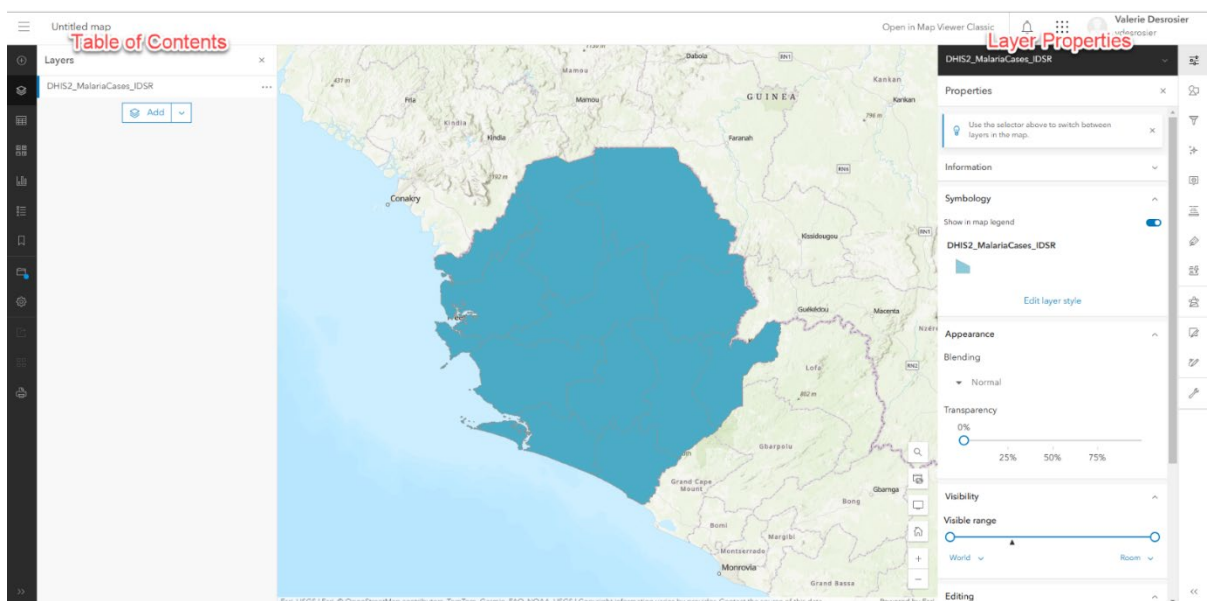
- This will open the **ArcGIS Enterprise organization**.

Visualize Data in ArcGIS Enterprise

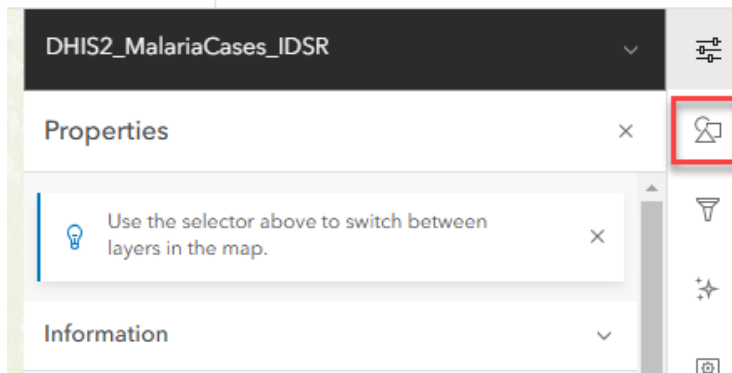
1. On the item page, select Open in Map Viewer.



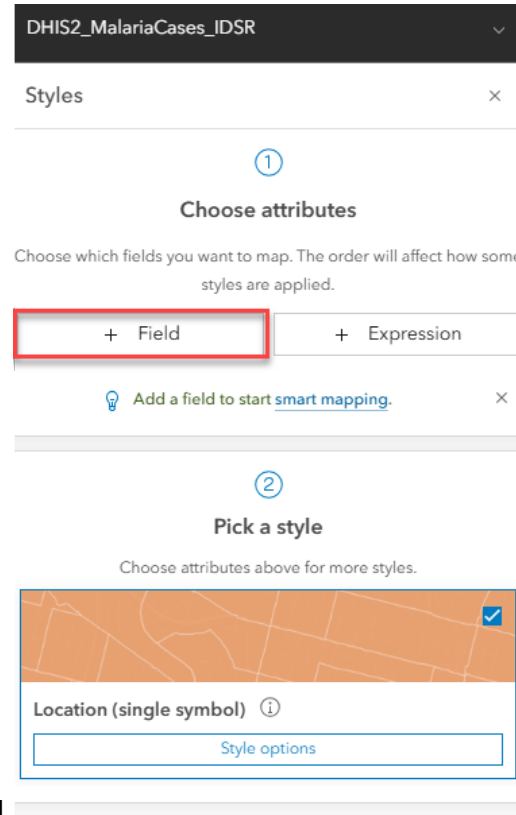
2. A new map will open and the properties of your new layer will be on the right side of the screen with the table of contents on the left.



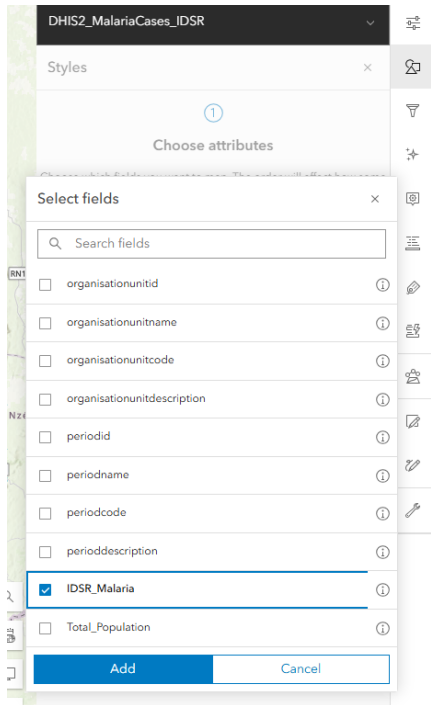
3. Open the **Styles** properties on the right to:



- Apply **Smart Mapping** for data visualization.
- Create **Choropleth Maps** based on values (e.g., malaria cases).
- Compare multiple datasets (e.g., Malaria Cases vs. Population).
- Use **relationship comparison** for dual dataset analysis.



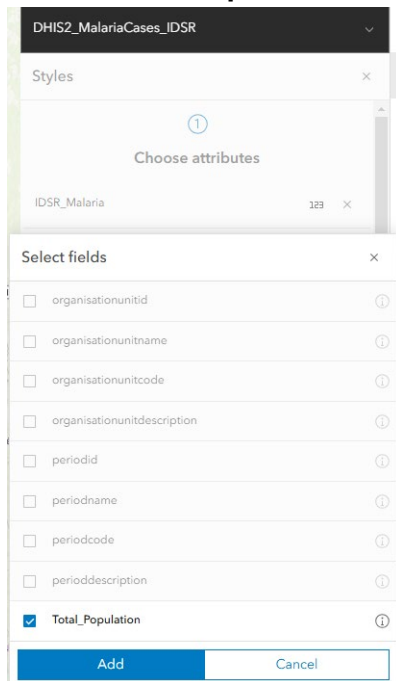
- Select **add Field**.
- Select **IDSR_Malaria**. Click **Add**.



6. Select Style Options to view different available mapping options.
7. To view symbolizing multiple data elements at once, select **add field** again.



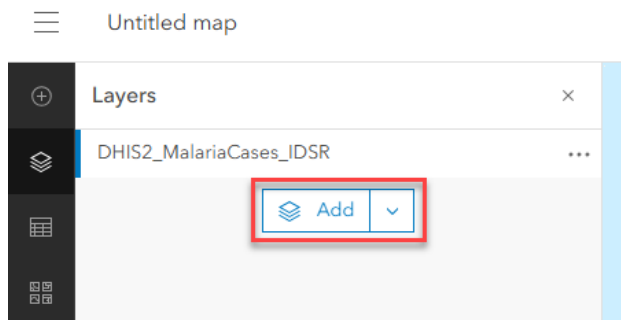
8. Select **Total_Population**. Click **Add**.



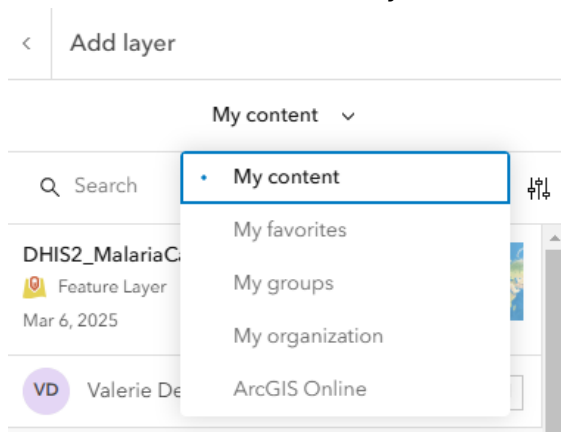
9. Explore the new symbol options.
10. To learn more, follow this tutorial on designing symbology in ArcGIS:
<https://learn.arcgis.com/en/projects/design-symbology-for-a-thematic-map/arcgis-online/>

Enhance your Analysis with Additional Data

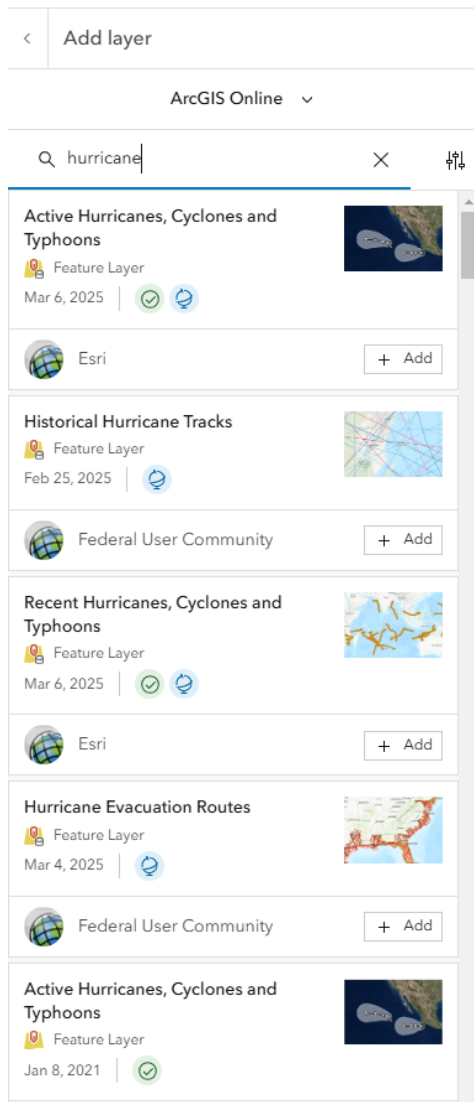
1. In your **Table of Contents**, select **Add**.



2. Browse for more of your own content to add to the map, content from your Organization, or content from **ArcGIS Online** and the **ArcGIS Living Atlas of the World** for additional data layers.



3. Add layers from sources such as:
 - a. ArcGIS Living Atlas (e.g., hurricanes, climate data).
 - b. NASA, NOAA, and other major organizations.



4. Explore what's available in the Living Atlas here:
<https://livingatlas.arcgis.com/en/home/>

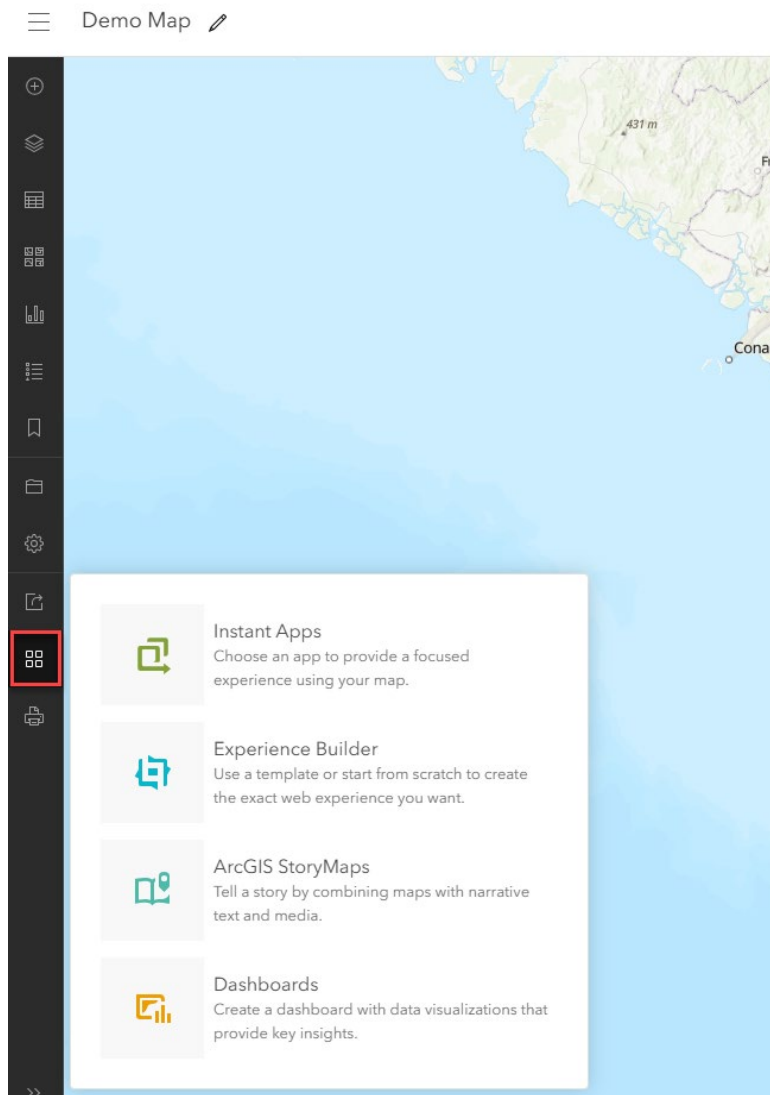
Perform Advanced Analytics

1. Use **ArcGIS analysis tools in your online map**  for:
 - a. Aggregating points.
 - b. Summarizing nearby locations.
 - c. Conducting **drive-time** or **walk-time** analysis.
 - d. Creating **buffer zones**.
 - e. Etc.

2. Save results for further visualization.
3. To get started with analysis, follow this tutorial on siting a new hospital location:
<https://learn.arcgis.com/en/projects/site-a-new-hospital/>

Create Web Applications from your DHIS2 Data

1. Save your map  and immediately use your map to create an app such as:
 - a. **Instant Apps** for simple sharing.
 - b. **Experience Builder** for interactive applications.
 - c. **Story Maps** for storytelling with spatial data.
 - d. **Dashboards** for real-time analytics.



- e. For training on dashboards, follow this tutorial to create an example disease dashboard: <https://learn.arcgis.com/en/projects/create-a-covid-19-dashboard/>

Next Steps and Additional Resources

For additional information on ArcGIS access, DHIS2 users can explore Esri's offerings [here](#).

Esri offers a variety of training resources to help users develop GIS skills, whether they are beginners or experienced professionals. Many training materials are available through [Esri Academy](#), which includes self-paced tutorials, instructor-led courses, and learning plans. However, most resources in Esri Academy require an **Esri account**, and some may require an organizational subscription or additional purchase.

For those looking for more industry specific learning opportunities, Esri also provides [tutorials](#), which typically walk users through hands-on exercises using Esri software. Like Esri Academy, these tutorials generally require an account and access to ArcGIS software.

For completely free learning that does not require an Esri account or software access, Esri offers **Massive Open Online Courses (MOOCs)** throughout the year. These MOOCs cover various GIS topics, such as spatial analysis, cartography, and imagery analysis. They are open to everyone, include interactive lessons, and provide temporary access to necessary Esri software during the course.

Suggested Training Opportunities:

- [Welcome to the ArcGIS System StoryMap](#)
- [Getting Started with ArcGIS Online](#)
- [Introduction to MapViewer](#)
- [Learn About Designing Symbolology in Maps](#)
- [Hands On Learning – Map a Historic Cholera Outbreak](#)
- [Get Started with Analysis – Site a New Hospital](#)
- [Using Analysis for Accessibility - Shelter Access in Hawaii](#)
- [Using Analysis for Accessibility – Food access in DC](#)
- [Create a Dashboard - Wildfire Tracking Example](#)
- [Create a Dashboard – COVID-19 Example](#)
- [Create a StoryMap](#)
- [Collect Data in the Field with Survey123](#)

- [Suitability Modeling in ArcGIS Pro](#)

Visit the [Health GIS Hub](#) for real-world examples.



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Founded in 1969 in Redlands, California, USA, Esri software is deployed in more than 350,000 organizations globally and in over 200,000 institutions in the Americas, Asia and the Pacific, Europe, Africa, and the Middle East. Esri has partners and local distributors in over 100 countries on six continents, including Fortune 500 companies, government agencies, nonprofits, and universities. With its pioneering commitment to geospatial information technology, Esri engineers the most innovative solutions for digital transformation, the Internet of Things (IoT), and advanced analytics.

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